



Department of Mathematics and Natural Sciences
MAT120- Mathematics II: Integral Calculus and Differential Equations
Graded Quiz 3
Total marks: 10
Time: 20 minutes

Instructions:

- Please write your name, ID, and section in your answer script.
- Answer all the questions.
- Submit the PDF of your answer script.

1. Find the general solution of the given DE: [5]

$$x \frac{dy}{dx} + 2y = \frac{\sin x}{x}$$

2. Determine whether the given DE is exact. If it is exact, then solve it. [5]

$$(y^2 e^{xy^2} + 4x^3) dx + (2xy e^{xy^2} - 3y^2) dy = 0$$